Question 1:

Let a be the y-intercept and θ be the angle the line makes with the x-axis.

$$L = L_{a,\theta} = \{(x, a + x\tan\theta) : x \in \mathbb{R}\}$$

and teh set of all such lines;

$$X = \{L_{a,\theta} : a \in \mathbb{R} \text{ and } \theta \in (\frac{-\pi}{2}, \frac{\pi}{2})\}$$

With distance function Δ on X;

$$\Delta(L_{a,\theta}, L_{b,\phi}) = \mid a - b \mid + \mid \tan\theta - \tan\phi \mid$$

a)

$$\begin{aligned}\Delta(L_{0,\frac{\pi}{4}}, L_{1,\frac{-\pi}{4}}) &= \mid 0 - 1 \mid + \mid \tan\frac{\pi}{4} - \tan\frac{-\pi}{4} \\ &= \mid 1 \mid + \mid 1 - (-1) \mid \\ &= 1 + 2 \\ &= 3\end{aligned}$$

b)
Using Definition 2.1 Chapter 14 p72;

i.
M1:

$$d(a,b) \geq 0$$

Let $L_{a,\theta}, L_{b,\phi} \in X$.

$$\begin{aligned}\Delta(L_{a,\theta}, L_{b,\phi}) &= \mid a - b \mid + \mid \tan\theta - \tan\phi \mid \\ &\geq 0 + 0 \\ &= 0\end{aligned}$$

Since absolute values are always non-negative, non-negativity holds.

Let $L_{a,\theta}, L_{b,\phi} \in X$.

$$\Delta(L_{a,\theta}, L_{b,\phi}) = \mid a - b \mid + \mid \tan\theta - \tan\phi \mid = 0$$

This holds if and only if both absolute value terms are zero

$$\begin{aligned}
&\iff \mid a - b \mid = 0 \text{ and } \mid \tan\theta - \tan\phi \mid = 0 \\
&\iff a = b \text{ and } \tan\theta = \tan\phi \\
&\iff a = b \text{ and } \theta = \phi \quad (\text{since } \theta, \phi \in (\frac{-\pi}{2}, \frac{\pi}{2})) \\
&\iff L_{a,\theta} = L_{b,\phi}
\end{aligned}$$

ii.
M2:

$$d(a,b) = d(b,a)$$

$$\begin{aligned}
\Delta(L_{a,\theta}, L_{b,\phi}) &= \mid a - b \mid + \mid \tan\theta - \tan\phi \mid \\
&= \mid b - a \mid + \mid \tan\phi - \tan\theta \mid \\
&= \Delta(L_{b,\phi}, L_{a,\theta})
\end{aligned}$$

Thus, symmetry holds.

iii.
M3:

$$d(a,c) \leq d(a,b) + d(b,c)$$

Let $L_{a,\theta}, L_{b,\phi}, L_{c,\psi} \in X$.

$$\begin{aligned}\Delta(L_{a,\theta}, L_{c,\psi}) &= \mid a - c \mid + \mid \tan\theta - \tan\psi \mid \\ &\leq \mid a - b \mid + \mid b - c \mid + \mid \tan\theta - \tan\phi \mid + \mid \tan\phi - \tan\psi \mid \\ &= \Delta(L_{a,\theta}, L_{b,\phi}) + \Delta(L_{b,\phi}, L_{c,\psi})\end{aligned}$$

Thus, the triangle inequality holds.

c)
Using Definition 2.6 Chapter 14 p72;

$$B_\Delta(L_{0,0}, 1)$$

$$\begin{aligned}B_\Delta(L_{0,0}, 1) &= \{L_{b,\phi} \in X : \Delta(L_{0,0}, L_{b,\phi}) < 1\} \\ &= \{L_{b,\phi} \in X : \mid 0 - b \mid + \mid \tan 0 - \tan\phi \mid < 1\} \\ &= \{L_{b,\phi} \in X : \mid b \mid + \mid 0 - \tan\phi \mid < 1\} \\ Applying the triangle inequality in \mathbb{R} \text{ to each absolute value term} &= \{L_{b,\phi} \in X : \mid b \mid + \mid \tan\phi \mid < 1\}\end{aligned}$$

d)
Is the sequence of lines given by;

$$(L_n)_{n=1}^{\infty}, L_n = L_{\frac{-1}{n}, \frac{\pi}{2} - \frac{1}{n}}$$

$\Delta - convergent$?

To be $\Delta - convergent$, there must exist some line $L_{b,\phi} \in X$ such that for every $\epsilon > 0$, there exists some $N \in \mathbb{N}$ such that for all $n \geq N$;

$$\Delta(L_n, L_{b,\phi}) < \epsilon$$

Calculating $\Delta(L_n, L_{b,\phi})$;

$$\begin{aligned}\Delta(L_n, L_{b,\phi}) &= \Delta(L_{\frac{-1}{n},\frac{\pi}{2}-\frac{1}{n}}, L_{b,\phi}) \\ &= \left| \frac{-1}{n} - b \right| + \left| \tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan\phi \right|\end{aligned}$$

Alternative

Question 2:

$$L = \mathbf{L_a}, \theta = \{(x, a + x\tan\theta) : x \in \mathbb{R}\}$$

Where a is the y- intercept of the line and θ is the angle of inclination of the line to the positive x-axis.

$$\mathbf{X} = \mathbf{L_a}, \theta : \mathbf{a} \in \mathbb{R} \text{ and } \theta \in (-\pi/2, \pi/2)$$

and

$$\Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) =\mid a - b \mid + \mid \tan\theta - \tan\phi \mid$$

for $\mathbf{L_a}, \theta, \mathbf{L_b}, \phi \in \mathbf{X}$

a)
The distance between $\mathbf{L_0}, \pi/4$ and $\mathbf{L_1}, -\pi/4$;

$$\begin{aligned}\Delta(\mathbf{L_0}, \pi/4, \mathbf{L_1}, -\pi/4) &=\mid 0 - 1 \mid + \mid \tan\pi/4 - \tan -\pi/4 \mid \\ &= 1 + \mid 1 - (-1) \mid \\ &= 1 + \mid 2 \mid \\ &= 1 + 2 = 3\end{aligned}$$

b)
Bt Definition 2.1,HB Chapter 14 p72,

i.
To show Δ satisfies (M1) non-negativity, we need to show that for any $d^{(2)}(\mathbf{a}, \mathbf{b}) \geq 0$

$$\Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) =\mid a - b \mid + \mid \tan\theta - \tan\phi \mid$$

As $\mathbf{a}, \mathbf{b} \in \mathbb{R}$ and $\theta \in (-\pi/2, \pi/2)$, both $\mid a - b \mid \geq 0$ and $\mid \tan\theta - \tan\phi \mid \geq 0$

$$\geq 0 + 0$$

Thus, $\Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) \geq 0$ and (M1) is satisfied.

ii.
To show Δ satisfies (M2) symetry, we need to show that for any $d^{(2)}(\mathbf{a}, \mathbf{b}) = d^{(2)}(\mathbf{b}, \mathbf{a})$

$$\Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) = \mid a - b \mid + \mid \tan\theta - \tan\phi \mid$$

By the properties of absolute values, HB p8, $\mid a - b \mid = \mid b - a \mid$ and $\mid \tan\theta - \tan\phi \mid = \mid \tan\phi - \tan\theta \mid$

$$= \mid b - a \mid + \mid \tan\phi - \tan\theta \mid$$

$$= \Delta(\mathbf{L_b}, \phi, \mathbf{L_a}, \theta)$$

Thus, $\Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) = \Delta(\mathbf{L_b}, \phi, \mathbf{L_a}, \theta)$ and (M2) is satisfied.

iii.
To show Δ satisfies (M3) triangle inequality, we need to show that for any $d^{(2)}(\mathbf{a}, \mathbf{c}) \leq d^{(2)}(\mathbf{a}, \mathbf{b}) + d^{(2)}(\mathbf{b}, \mathbf{c})$

$$\Delta(\mathbf{L_a}, \theta, \mathbf{L_c}, \psi) = \mid a - c \mid + \mid \tan\theta - \tan\psi \mid$$

By the triangle inequality for real numbers, HB p9, $\mid a - c \mid \leq \mid a - b \mid + \mid b - c \mid$ and $\mid \tan\theta - \tan\psi \mid \leq \mid \tan\theta - \tan\phi \mid + \mid \tan\phi - \tan\psi \mid$

$$\leq (\mid a - b \mid + \mid b - c \mid) + (\mid \tan\theta - \tan\phi \mid + \mid \tan\phi - \tan\psi \mid)$$

$$= \mid a - b \mid + \mid \tan\theta - \tan\phi \mid + \mid b - c \mid + \mid \tan\phi - \tan\psi \mid$$

$$= \Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) + \Delta(\mathbf{L_b}, \phi, \mathbf{L_c}, \psi)$$

Thus, $\Delta(\mathbf{L_a}, \theta, \mathbf{L_c}, \psi) \leq \Delta(\mathbf{L_a}, \theta, \mathbf{L_b}, \phi) + \Delta(\mathbf{L_b}, \phi, \mathbf{L_c}, \psi)$ and (M3) is satisfied.

c)

$$\mathbf{B_{\Delta}}(\mathbf{L_{0,0}}, 1)$$

By Definition 2.6, HB Chapter 14 p72,

$$\begin{aligned}
\mathbf{B_{\Delta}}(\mathbf{L_{0,0}}, 1) &= \{\mathbf{L_b}, \theta \in \mathbf{X} : \Delta(\mathbf{L_{0,0}}, \mathbf{L_b}, \theta) < 1\} \\
&= \{\mathbf{L_b}, \theta \in \mathbf{X} : \mid 0 - b \mid + \mid \tan 0 - \tan\theta \mid < 1\} \\
&= \{\mathbf{L_b}, \theta \in \mathbf{X} : \mid b \mid + \mid 0 - \tan\theta \mid < 1\} \\
&= \{\mathbf{L_b}, \theta \in \mathbf{X} : \mid b \mid + \mid \tan\theta \mid < 1\}
\end{aligned}$$

d)

The sequence of lines;

$$(\mathbf{L_n})_{n=1}^{\infty} \text{ given by } \mathbf{L_n} = \mathbf{L}_{-1/n, \pi/2 - 1/n}$$

is not $\Delta - convergent$ because, by Definition 3.2 HB Chapter 14 p73;

Using $L_{b,\phi} \in X$ Then

$$\Delta(L_n, L_{b,\phi}) = \left|-\frac{1}{n} - b\right| + \left|\tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan\phi\right|$$

As $n \to \infty$, we have $-1/n \to 0$, so

$$\left|-\frac{1}{n} - b\right| \to |b|$$

Also, as $n \to \infty$, we have

$$\tan\left(\frac{\pi}{2} - \frac{1}{n}\right) \to +\infty$$

and since $\tan\phi$ is finite for all $\phi \in (-\pi/2, \pi/2)$,

$$\left|\tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan\phi\right| \to \infty.$$

Hence $\Delta(L_n, L_{b,\phi}) \to \infty$, and so (L_n) is not Δ-convergent.